

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-11 22:30 UTC

SEC P vs NP — notebook summary

Generated 2026-05-11 22:30 UTC

- Cycles recorded: **627**
- Time span: 432.8h (~1.45 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	590
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	34
Schur-Weyl Duality	21
Representation Theory of Symmetric Groups	17
Free Probability	14
Matroid Theory	13
Additive Combinatorics	10
Free Probability Theory	8
Noncommutative L^p Geometry	8
Polymatroid Theory	8
Algebraic Geometry	7
Random Matrix Theory	7
Finite Geometry	6
Spectral Graph Theory	6
Plethysm Theory	6
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4

Field	Cycles
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Diophantine Approximation	3
Persistent Homology	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-11 15:49 UTC	INCONCLUSIVE	Plethysm Coefficient Gap in Monotone Circuit Complexity for Perma
2026-05-11 16:11 UTC	INCONCLUSIVE	Real Rank of Depth-3 AC ⁰ Circuits for PARITY is $\Omega(\log n)$
2026-05-11 17:00 UTC	INCONCLUSIVE	Quadratic Form Solution Count Exponential Gap in ACC ⁰ Sipser Circ
2026-05-11 17:52 UTC	INCONCLUSIVE	Additive Energy Inverse Proportional to Discrepancy of Sumset Com
2026-05-11 18:06 UTC	INCONCLUSIVE	Symmetric Square Multiplicity Gap in Permanent vs Determinant Dec
2026-05-11 18:31 UTC	INCONCLUSIVE	Free Cumulant Sum Gap in Read-Twice BP Transition Matrices
2026-05-11 19:29 UTC	INCONCLUSIVE	Colin de Verdière Invariant Lower Bounds Resolution Length for Ts
2026-05-11 19:40 UTC	INCONCLUSIVE	Free Cumulant Gap in Read-Twice BPs for IP ₂
2026-05-11 20:25 UTC	INCONCLUSIVE	Free Cumulant Sum Gap in Read-Twice BPs for IP ₂
2026-05-11 20:56 UTC	INCONCLUSIVE	Free Cumulant Gap in Read-Twice BPs for IP ₂
2026-05-11 21:18 UTC	INCONCLUSIVE	Tropical Convex Hull Dimension Inverse Proportional to ACC ⁰ Circ
2026-05-11 21:40 UTC	INCONCLUSIVE	Symmetric Power Multiplicity Gap in Permanent vs Determinant Repr

Time	Verdict	Title
2026-05-11 21:47 UTC	INCONCLUSIVE	Schur-Weyl Multiplicity Gap in Permanent vs Determinant Decomposi
2026-05-11 21:57 UTC	INCONCLUSIVE	Dual Convex Body Width Bounds SOS Degree for Max-CUT
2026-05-11 22:27 UTC	INCONCLUSIVE	Cycle Matroid Rank Inverse Proportional to Resolution Proof Size

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```